Appendix 3. Database search strings

Searches conducted 18 May 2020:

PubMed

(("Masks/adverse effects"[MAJR])

OR

((Masks[Mesh] OR "Respiratory protective devices"[Mesh] OR Facemask[tiab] OR Facemasks[tiab] OR Mask[tiab] OR Masks[tiab] OR Faceshield[tiab] OR Faceshields[tiab] OR Goggles[tiab] OR “Face shield”[tiab] OR “Face shields”[tiab] OR Visors[tiab] OR “Face covering”[tiab])

AND

("Adverse effects"[sh] OR “Adverse effect”[tiab] OR “Adverse effects”[tiab] OR “Adversely affect”[tiab] OR “Adverse event”[tiab] OR “Adverse events”[tiab] OR Discomfort[tiab] OR Fatigue[tiab] OR Pain[tiab] OR “Difficulty breathing”[tiab] OR “Breathing difficulty”[tiab] OR “Shortness of breath”[tiab] OR “Risk compensation”[tiab] OR Fomite[tiab] OR Fomites[tiab] OR Dislike[tiab] OR “Physiological impacts”[tiab] OR “Physiological impact”[tiab] OR Sweating[tiab] OR Irritation[tiab] OR “Psychological reactions”[tiab] OR Fear[tiab] OR Communication[tiab] OR “Deaf persons”[tiab] OR Lip-reading[tiab] OR “Lip reading”[tiab] OR Vitiligo[tiab] OR Humidity[tiab] OR “Face dermatitis”[tiab] OR Acne[tiab] OR Redness[tiab] OR Ulcer[tiab] OR “Speech understanding”[tiab] OR “Pinching of skin”[tiab] OR “Moisture buildup”[tiab] OR “Facial warmth”[tiab])

AND

(Relieve[tiab] OR Attitudes[tiab] OR Induced[tiab] OR Modified[tiab] OR Decreased[tiab] OR Decrease[tiab] OR Associated[tiab] OR Impair[tiab] OR Reduce[tiab] OR Reducing[tiab] OR Reduced[tiab] OR Duration[tiab])

AND

(“randomized controlled trial”[pt] OR “controlled clinical trial”[pt] OR randomized[tiab] OR randomised[tiab] OR placebo[tiab] OR randomly[tiab] OR trial[tiab] OR groups[tiab] OR "Epidemiologic Studies"[Mesh] OR “case-control studies”[Mesh] OR “Cohort Studies”[Mesh] OR “case control”[tiab] OR Cohort[tiab] OR “Follow up”[tiab] OR Observational[tiab] OR Longitudinal[tiab] OR Prospective[tiab] OR Retrospective[tiab] OR “cross sectional”[tiab] OR “Cross-Sectional Studies”[Mesh] OR Evaluated[tiab] OR Impact[tiab] OR Analysis[tiab] OR Statistics[tiab] OR Data[tiab] OR "statistics and numerical data"[sh] OR "epidemiology"[sh] OR Etiology[sh] OR Effect[tiab] OR Effects[tiab] OR Affect[tiab])))

NOT

("Continuous Positive Airway Pressure"[Mesh] OR "Positive Airway Pressure"[tiab] OR CPAP[tiab] OR "Laryngeal Masks"[Mesh] OR "Intubation, Intratracheal"[Mesh] OR “Oxygen masks”[tiab] OR

"Oxygen mask"[tiab] OR OxyMask[tiab] OR "Nasal masks"[tiab] OR "Patil-Syracuse Mask"[tiab] OR "Ventilation masks"[tiab] OR "Venti-Mask"[tiab] OR "Venturi mask"[tiab] OR "Diving masks"[tiab] OR "Thermoplastic masks"[tiab] OR Bag-valve-mask[tiab] OR "Laryngeal mask"[tiab] OR "Laryngeal masks"[tiab] OR "Oronasal masks"[tiab] OR "Sleep mask"[tiab] OR Editorial[pt] OR Letter[pt] OR "Case reports"[pt])

NOT

(Animals[Mesh] not (Animals[Mesh] and Humans[Mesh]))

OR

(Masks[Mesh] OR "Respiratory protective devices"[Mesh] OR Facemask[tiab] OR Facemasks[tiab] OR Mask[tiab] OR Masks[tiab] OR Faceshield[tiab] OR Faceshields[tiab] OR Goggles[tiab] OR "Face shield"[tiab] OR "Face shields"[tiab] OR Visors[tiab] OR "Face covering"[tiab])

AND

("Respiratory Tract Infections"[Mesh] OR Influenza[tiab] OR "Respiratory Tract Diseases"[Mesh] OR Influenzas[tiab] OR "Influenza-like"[tiab] OR ILI[tiab] OR "common cold"[tiab] OR colds[tiab] OR coryza[tiab] OR coronavirus[Mesh] OR "sars virus"[Mesh] OR coronavirus[tiab] OR Coronaviruses[tiab] OR "coronavirus infections"[Mesh] OR "severe acute respiratory syndrome"[Mesh] OR "severe acute respiratory syndrome"[tiab] OR "severe acute respiratory syndromes"[tiab] OR sars[tiab] OR "respiratory syncytial viruses"[Mesh] OR "respiratory syncytial virus, human"[Mesh] OR "Respiratory Syncytial Virus Infections"[Mesh] OR "respiratory syncytial virus"[tiab] OR "respiratory syncytial viruses"[tiab] OR rsv[tiab] OR parainfluenza[tiab] OR "Respiratory illness"[tiab] OR ((Transmission[tiab]) AND (Coughing[tiab] OR Sneezing[tiab])) OR ((respiratory[tiab] AND Tract[tiab]) AND (infection[tiab] OR Infections[tiab] OR illness[tiab])) OR "Communicable Disease Control"[Mesh] OR "Disease Outbreaks"[Mesh] OR "Disease Transmission, Infectious"[Mesh] OR "Infection Control"[Mesh] OR Transmission[sh] OR "Communicable Disease Control"[tiab] OR "Secondary transmission"[tiab] OR ((Reduced[tiab] OR Reduce[tiab] OR Reduction[tiab] OR Reducing[tiab] OR Lower[tiab]) AND (Incidence[tiab] OR Occurrence[tiab] OR Transmission[tiab] OR Secondary[tiab])) OR Physiological[tiab])

AND

("Adverse effects"[sh] OR "Adverse effect"[tiab] OR "Adverse effects"[tiab] OR "Adverse event"[tiab] OR "Adverse events"[tiab] OR "Index cases"[tiab] OR "Attack rates"[tiab] OR ((Reduce[tiab] OR Reduction[tiab]) AND (Rate[tiab] OR Rates[tiab])) OR Perception[tiab] OR Prevent[tiab] OR Adherence[tiab])

AND

("randomized controlled trial"[pt] OR "controlled clinical trial"[pt] OR randomized[tiab] OR randomised[tiab] OR placebo[tiab] OR randomly[tiab] OR trial[tiab] OR groups[tiab] OR "Epidemiologic Studies"[Mesh] OR "case-control studies"[Mesh] OR "Cohort Studies"[Mesh] OR "case control"[tiab] OR Cohort[tiab] OR "Follow up"[tiab] OR Observational[tiab] OR Longitudinal[tiab] OR Prospective[tiab] OR retrospective[tiab] OR "cross sectional"[tiab] OR "Cross-

Sectional Studies”[Mesh] OR Evaluated[tiab] OR Impact[tiab] OR Analysis[tiab] OR Statistics[tiab] OR Data[tiab] OR "statistics and numerical data"[sh] OR "epidemiology"[sh])

NOT

("Continuous Positive Airway Pressure"[Mesh] OR "Positive Airway Pressure"[tiab] OR CPAP[tiab] OR "Laryngeal Masks"[Mesh] OR "Intubation, Intratracheal"[Mesh] OR “Oxygen masks”[tiab] OR “Oxygen mask”[tiab] OR OxyMask[tiab] OR “Goldman nasal masks”[tiab] OR “Patil-Syracuse Mask”[tiab] OR “Ventilation masks”[tiab] OR “Venti-Mask”[tiab] OR “Venturi mask”[tiab] OR “Hockey mask”[tiab] OR “Diving masks”[tiab] OR “Thermoplastic masks”[tiab] OR Bag-valve-mask[tiab] OR “Laryngeal mask”[tiab] OR “Laryngeal masks”[tiab] OR “Oronasal masks”[tiab] OR “Sleep mask”[tiab] OR "Case Reports"[pt])

NOT

(Animals[Mesh] not (Animals[Mesh] and Humans[Mesh]))

Cochrane CENTRAL

(([mh Masks/AE])

OR

(([mh Masks] OR [mh "Respiratory protective devices"] OR Facemask:ti,ab OR Facemasks:ti,ab OR Mask:ti,ab OR Masks:ti,ab OR Faceshield:ti,ab OR Faceshields:ti,ab OR Goggles:ti,ab OR "Face shield":ti,ab OR "Face shields":ti,ab OR Visors:ti,ab OR "Face covering":ti,ab)

AND

("Adverse effect":ti,ab OR "Adverse effects":ti,ab OR "Adversely affect":ti,ab OR "Adverse event":ti,ab OR "Adverse events":ti,ab OR Discomfort:ti,ab OR Fatigue:ti,ab OR Pain:ti,ab OR "Difficulty breathing":ti,ab OR "Breathing difficulty":ti,ab OR "Shortness of breath":ti,ab OR "Risk compensation":ti,ab OR Fomite:ti,ab OR Fomites:ti,ab OR Dislike:ti,ab OR "Physiological impacts":ti,ab OR "Physiological impact":ti,ab OR Sweating:ti,ab OR Irritation:ti,ab OR "Psychological reactions":ti,ab OR Fear:ti,ab OR Communication:ti,ab OR "Deaf persons":ti,ab OR "Lip reading":ti,ab OR Vitiligo:ti,ab OR Humidity:ti,ab OR "Glasses misting" OR "Face dermatitis":ti,ab OR Acne:ti,ab OR Redness:ti,ab OR Ulcer:ti,ab OR "Speech understanding":ti,ab OR "Pinching of skin":ti,ab OR "Moisture buildup":ti,ab OR "Facial warmth":ti,ab)

AND

(Relieve:ti,ab OR Attitudes:ti,ab OR Induced:ti,ab OR Modified:ti,ab OR Decreased:ti,ab OR Decrease:ti,ab OR Associated:ti,ab OR Impair:ti,ab OR Reduce:ti,ab OR Reducing:ti,ab OR Reduced:ti,ab OR Duration:ti,ab)))

NOT

([mh "Continuous Positive Airway Pressure"] OR "Positive Airway Pressure":ti,ab OR CPAP:ti,ab OR [mh "Laryngeal Masks"] OR [mh "Intubation, Intratracheal"] OR "Oxygen masks":ti,ab OR "Oxygen mask":ti,ab OR OxyMask:ti,ab OR "Nasal masks":ti,ab OR "Patil Syracuse Mask":ti,ab OR "Ventilation masks":ti,ab OR “Venti Mask”:ti,ab OR "Venturi mask":ti,ab OR "Hockey mask":ti,ab OR "Diving masks":ti,ab OR "Thermoplastic masks":ti,ab OR “Bag valve mask”:ti,ab OR "Laryngeal mask":ti,ab OR "Laryngeal masks":ti,ab OR "Oronasal masks":ti,ab OR "Sleep mask":ti,ab)

OR

([mh Masks] OR [mh "Respiratory protective devices"] OR Facemask:ti,ab OR Facemasks:ti,ab OR Mask:ti,ab OR Masks:ti,ab OR Faceshield:ti,ab OR Faceshields:ti,ab OR Goggles:ti,ab OR "Face shield":ti,ab OR "Face shields":ti,ab OR Visors:ti,ab OR "Face covering":ti,ab)

AND

([mh "Respiratory Tract Infections"] OR Influenza:ti,ab OR [mh "Respiratory Tract Diseases"] OR Influenzas:ti,ab OR ILI:ti,ab OR "common cold":ti,ab OR colds:ti,ab OR coryza:ti,ab OR [mh coronavirus] OR [mh "sars virus"] OR coronavirus:ti,ab OR Coronaviruses:ti,ab OR [mh "coronavirus infections"] OR [mh "severe acute respiratory syndrome"] OR "severe acute respiratory syndrome":ti,ab OR "severe acute respiratory syndromes":ti,ab OR sars:ti,ab OR [mh "respiratory syncytial viruses"] OR [mh "respiratory syncytial virus, human"] OR [mh "Respiratory Syncytial Virus Infections"] OR "respiratory syncytial virus":ti,ab OR "respiratory syncytial viruses":ti,ab OR rsv:ti,ab OR parainfluenza:ti,ab OR "Respiratory illness":ti,ab OR ((Transmission:ti,ab) AND (Coughing:ti,ab OR Sneezing:ti,ab)) OR ((respiratory:ti,ab AND Tract:ti,ab) AND (infection:ti,ab OR Infections:ti,ab OR illness:ti,ab)) OR [mh "Communicable Disease Control"] OR [mh "Disease Outbreaks"] OR [mh "Disease Transmission, Infectious"] OR [mh "Infection Control"] OR "Communicable Disease Control":ti,ab OR "Secondary transmission":ti,ab OR ((Reduced:ti,ab OR Reduce:ti,ab OR Reduction:ti,ab OR Reducing:ti,ab OR Lower:ti,ab) AND (Incidence:ti,ab OR Occurrence:ti,ab OR Transmission:ti,ab OR Secondary:ti,ab)) OR Physiological:ti,ab)

AND

("Adverse effect":ti,ab OR "Adverse effects":ti,ab OR "Adverse event":ti,ab OR "Adverse events":ti,ab OR "Index cases":ti,ab OR "Attack rates":ti,ab OR ((Reduce:ti,ab OR Reduction:ti,ab) AND (Rate:ti,ab OR Rates:ti,ab)) OR Perception:ti,ab OR Prevent:ti,ab OR Adherence:ti,ab)

NOT

([mh "Continuous Positive Airway Pressure"] OR "Positive Airway Pressure":ti,ab OR CPAP:ti,ab OR [mh "Laryngeal Masks"] OR [mh "Intubation, Intratracheal"] OR "Oxygen masks":ti,ab OR "Oxygen mask":ti,ab OR OxyMask:ti,ab OR "Goldman nasal masks":ti,ab OR "Patil-Syracuse Mask":ti,ab OR "Ventilation masks":ti,ab OR “Venti Mask”:ti,ab OR "Venturi mask":ti,ab OR "Hockey mask":ti,ab OR "Diving masks":ti,ab OR "Thermoplastic masks":ti,ab OR “Bag valve mask”:ti,ab OR "Laryngeal mask":ti,ab OR "Laryngeal masks":ti,ab OR "Oronasal masks":ti,ab OR "Sleep mask":ti,ab)

Embase

('Masks'/exp OR 'Respiratory protective devices'/exp OR Facemask:ti,ab OR Facemasks:ti,ab OR Mask:ti,ab OR Masks:ti,ab OR Faceshield:ti,ab OR Faceshields:ti,ab OR Goggles:ti,ab OR "Face shield":ti,ab OR "Face shields":ti,ab OR Visors:ti,ab OR "Face covering":ti,ab)

AND

("Adverse effect":ti,ab OR "Adverse effects":ti,ab OR "Adversely affect":ti,ab OR "Adverse event":ti,ab OR "Adverse events":ti,ab OR Discomfort:ti,ab OR Fatigue:ti,ab OR Pain:ti,ab OR "Difficulty breathing":ti,ab OR "Breathing difficulty":ti,ab OR "Shortness of breath":ti,ab OR "Risk compensation":ti,ab OR Fomite:ti,ab OR Fomites:ti,ab OR Dislike:ti,ab OR "Physiological impacts":ti,ab OR "Physiological impact":ti,ab OR Sweating:ti,ab OR Irritation:ti,ab OR "Psychological reactions":ti,ab OR Fear:ti,ab OR Communication:ti,ab OR "Deaf persons":ti,ab OR Lip-reading:ti,ab OR "Lip reading":ti,ab OR Vitiligo:ti,ab OR Humidity:ti,ab OR "Glasses misting" OR "Face dermatitis":ti,ab OR Acne:ti,ab OR Redness:ti,ab OR Ulcer:ti,ab OR "Speech understanding":ti,ab OR "Pinching of skin":ti,ab OR "Moisture buildup":ti,ab OR "Facial warmth":ti,ab)

AND

(Relieve:ti,ab OR Attitudes:ti,ab OR Induced:ti,ab OR Modified:ti,ab OR Decreased:ti,ab OR Decrease:ti,ab OR Associated:ti,ab OR Impair:ti,ab OR Reduce:ti,ab OR Reducing:ti,ab OR Reduced:ti,ab OR Duration:ti,ab)

AND

("randomized controlled trial":it OR "controlled clinical trial":it OR randomized:ti,ab OR randomised:ti,ab OR placebo:ti,ab OR randomly:ti,ab OR trial:ti,ab OR groups:ti,ab OR 'Epidemiology'/exp OR 'case control study'/exp OR 'cohort analysis'/exp OR "case control":ti,ab OR Cohort:ti,ab OR "Follow up":ti,ab OR Observational:ti,ab OR Longitudinal:ti,ab OR Prospective:ti,ab OR retrospective:ti,ab OR "cross sectional":ti,ab OR 'cross-sectional study'/exp OR Evaluated:ti,ab OR Impact:ti,ab OR Analysis:ti,ab OR Statistics:ti,ab OR Data:ti,ab OR Etiology:ti,ab OR Effect:ti,ab OR Effects:ti,ab OR Affect:ti,ab)

NOT

('positive end expiratory pressure'/exp OR "Positive Airway Pressure":ti,ab OR CPAP:ti,ab OR 'laryngeal mask'/exp OR 'endotracheal intubation'/exp OR "Oxygen masks":ti,ab OR "Oxygen mask":ti,ab OR OxyMask:ti,ab OR "Goldman nasal masks":ti,ab OR "Patil-Syracuse Mask":ti,ab OR "Ventilation masks":ti,ab OR Venti-Mask:ti,ab OR "Venturi mask":ti,ab OR "Hockey mask":ti,ab OR "Diving masks":ti,ab OR "Thermoplastic masks":ti,ab OR Bag-valve-mask:ti,ab OR "Laryngeal mask":ti,ab OR "Laryngeal masks":ti,ab OR "Oronasal masks":ti,ab OR "Sleep mask":ti,ab OR 'letter'/exp OR 'editorial'/exp OR 'case report'/exp)

NOT

('Animal'/exp NOT ('Animal'/exp AND 'Human'/exp))

AND

[embase]/lim

OR

('Mask'/exp OR 'gas mask'/exp OR Facemask:ti,ab OR Facemasks:ti,ab OR Mask:ti,ab OR Masks:ti,ab OR Faceshield:ti,ab OR Faceshields:ti,ab OR Goggles:ti,ab OR "Face shield":ti,ab OR "Face shields":ti,ab OR Visors:ti,ab OR "Face covering":ti,ab)

AND

('respiratory tract infection'/exp OR Influenza:ti,ab OR 'respiratory tract disease'/exp OR Influenzas:ti,ab OR Influenza-like:ti,ab OR ILI:ti,ab OR "common cold":ti,ab OR colds:ti,ab OR coryza:ti,ab OR 'Coronavirinae'/exp OR 'SARS coronavirus'/exp OR coronavirus:ti,ab OR Coronaviruses:ti,ab OR 'Coronavirus infection'/exp OR 'severe acute respiratory syndrome'/exp OR "severe acute respiratory syndrome":ti,ab OR "severe acute respiratory syndromes":ti,ab OR sars:ti,ab OR 'Pneumovirus'/exp OR 'Human respiratory syncytial virus'/exp OR 'respiratory syncytial virus infection'/exp OR "respiratory syncytial virus":ti,ab OR "respiratory syncytial viruses":ti,ab OR rsv:ti,ab OR parainfluenza:ti,ab OR "Respiratory illness":ti,ab OR ((Transmission:ti,ab) AND (Coughing:ti,ab OR Sneezing:ti,ab)) OR ((respiratory:ti,ab AND Tract:ti,ab) AND (infection:ti,ab OR Infections:ti,ab OR illness:ti,ab)) OR 'communicable disease control'/exp OR 'epidemic'/exp OR 'disease transmission'/exp OR 'Infection Control'/exp OR "Communicable Disease Control":ti,ab OR "Secondary transmission":ti,ab OR ((Reduced:ti,ab OR Reduce:ti,ab OR Reduction:ti,ab OR Reducing:ti,ab OR Lower:ti,ab) AND (Incidence:ti,ab OR Occurrence:ti,ab OR Transmission:ti,ab OR Secondary:ti,ab)) OR Physiological:ti,ab)

AND

('adverse event'/exp OR "Adverse effect":ti,ab OR "Adverse effects":ti,ab OR "Adverse event":ti,ab OR "Adverse events":ti,ab OR "Index cases":ti,ab OR "Attack rates":ti,ab OR ((Reduce:ti,ab OR Reduction:ti,ab) AND (Rate:ti,ab OR Rates:ti,ab)) OR Perception:ti,ab OR Prevent:ti,ab OR Adherence:ti,ab)

AND

("randomized controlled trial":it OR "controlled clinical trial":it OR randomized:ti,ab OR randomised:ti,ab OR placebo:ti,ab OR randomly:ti,ab OR trial:ti,ab OR groups:ti,ab OR 'Epidemiology'/exp OR 'case control study'/exp OR 'cohort analysis'/exp OR "case control":ti,ab OR Cohort:ti,ab OR "Follow up":ti,ab OR Observational:ti,ab OR Longitudinal:ti,ab OR Prospective:ti,ab OR retrospective:ti,ab OR "cross sectional":ti,ab OR 'cross-sectional study'/exp OR Evaluated:ti,ab OR Impact:ti,ab OR Analysis:ti,ab OR Statistics:ti,ab OR Data:ti,ab)

NOT

('positive end expiratory pressure'/exp OR "Positive Airway Pressure":ti,ab OR CPAP:ti,ab OR 'laryngeal mask'/exp OR 'endotracheal intubation'/exp OR "Oxygen masks":ti,ab OR "Oxygen mask":ti,ab OR OxyMask:ti,ab OR "Goldman nasal masks":ti,ab OR "Patil-Syracuse Mask":ti,ab OR "Ventilation masks":ti,ab OR Venti-Mask:ti,ab OR "Venturi mask":ti,ab OR "Hockey mask":ti,ab OR "Diving masks":ti,ab OR "Thermoplastic masks":ti,ab OR Bag-valve-mask:ti,ab OR "Laryngeal

mask":ti,ab OR "Laryngeal masks":ti,ab OR "Oronasal masks":ti,ab OR "Sleep mask":ti,ab OR "Case Reports":it)

NOT

('Animal'/exp NOT ('Animal'/exp AND 'Human'/exp))

AND

[embase]/lim

Europe PMC

https://europepmc.org/search?query=%28TITLE%3A”masks”%20AND%20%28%222019-nCoV%22%20OR%20%222019nCoV%22%20OR%20%22COVID-19%22%20OR%20%22SARS-CoV-2%22%20OR%20%28%22wuhan%22%20AND%20%22coronavirus%22%29%20OR%20%22Coronavirus%22%20OR%20%22Corona%20virus%22%20OR%20%22corona-virus%22%20OR%20%22corona%20viruses%22%20OR%20%22coronaviruses%22%20OR%20%22SARS-CoV%22%20OR%20%22Orthocoronavirinae%22%20OR%20%22MERS-CoV%22%20OR%20%22Severe%20Acute%20Respiratory%20Syndrome%22%20OR%20%22Middle%20East%20Respiratory%20Syndrome%22%20OR%20%28%22SARS%22%20AND%20%22virus%22%29%20OR%20%22soluble%20ACE2%22%20OR%20%28%22ACE2%22%20AND%20%22virus%22%29%20OR%20%28%22ARDS%22%20AND%20%22virus%22%29%20or%20%28%22angiotensin-converting%20enzyme%202%22%20AND%20%22virus%22%29%29%29%20AND%20%28SRC%3APPR%29&page=1